RV College of Engineering®

ELEMENTS OF MECHANICAL ENGINEERING (22ME13/23)

Unit- V
MECHATRONICS & ROBOTICS

Presented by

Dr. Gangadhar Angadi
Department of Mechanical Engineering,
RV College of Engineering
Bangaluru-560059

Go, change the world®

RV College of Engineering®

Unit- V
Mechatronics & Robotics

Go, change the world®

Introduction: Evolution of Mechatronic system, measurement & control system, basic elements of control system, Applications-water level controller, washing machine, Engine management system (EMS), Anti-lock Braking System (ABS).

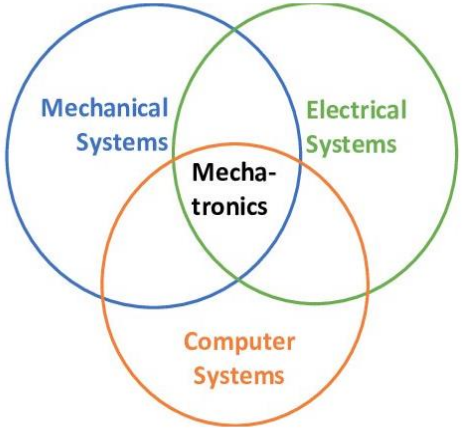
Robotics: Basic Structure of Robots, Robot Anatomy, Complete Classification of Robots, Fundamentals about Robot Technology, Basic Robot Configurations and their Relative Merits and Demerits.

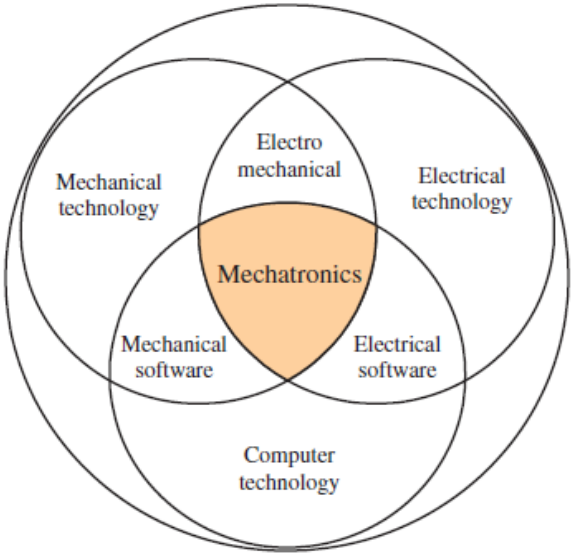
Introduction

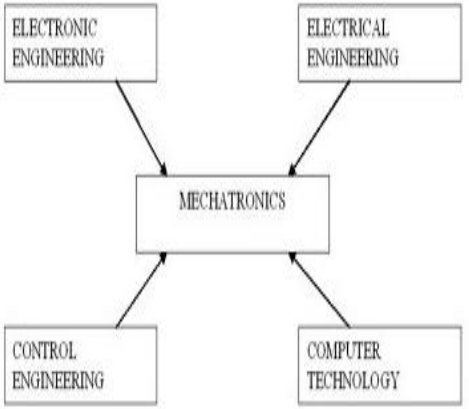
The term "**Mechatronics** " was coined in 1969 by a senior engineer of a Japanese Yasakawa Electric Company to refer to the use of electronics in mechanical control.

Mechatronics is defined as the synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of products and process

Domain of Mechatronics







Domain of Mechatronics

Mechatronics is defined as the synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of products and process.

Mechanical System - It consists of the moving parts of the system like drives, pulleys, gears, links and their mechanism.

Electrical System : It comprises of Microcontrollers, analog-digital and digital-analog converters, sensors, actuators, circuits, etc.

Computer System - Software, microprocessor and programmable logic controllers comes under this.

Electro-mechanical – It combines electrical and mechanical components to achieve integrated functionalities.

Mechanical Software – These are the softwares designed to control and optimize mechanical components which involves algorithms for automation.

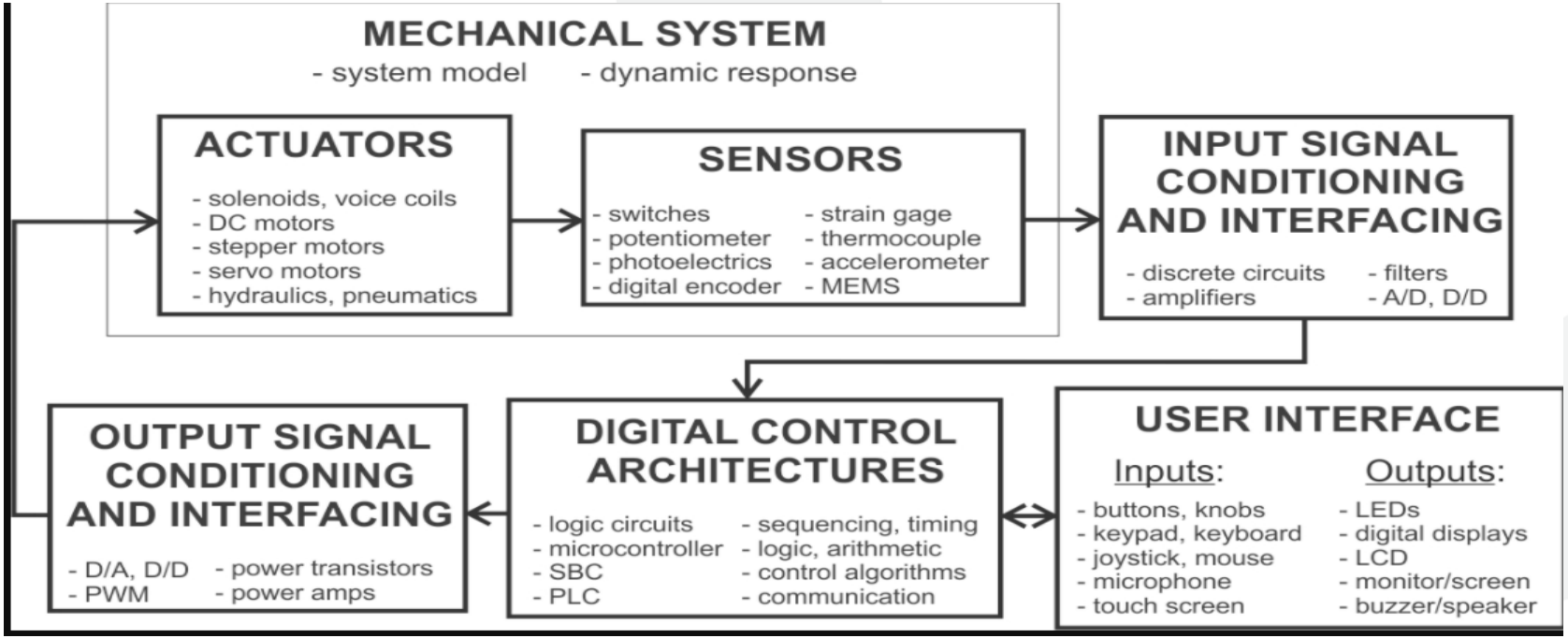
Electro-software – They focus on managing and controlling electronic components, contributing to the integration of electronics with software systems.

Hence, mechatronics is a multidisciplinary field that combines the principles of mechanical, electrical and computer engineering, integrating them to design and create intelligent systems with enhanced

RV College of Engineering®

Key elements of Mechatronics systems

Go, change the world®



https://youtu.be/Ro_tFv1iH6g

Key elements of Mechatronics systems

Actuators in mechatronic systems are devices that convert electrical signals into mechanical motion.

Here's an overview of the mentioned types of actuators:

1. **Solenoids, Voice Coils:** Convert electrical energy into linear motion (solenoids)
Application: Used in car door locks, fuel injectors, and valves in the engine control system.

2. **DC Motors:** Convert electrical energy into rotary motion.
Application: Powering electric windows, windshield wipers, and seat adjustments.

3. **Stepper Motors:** Convert electrical pulses into discrete, step-by-step rotational motion.
Application: Control of air conditioning vents, throttle control in drive-by-wire systems.

4. **Servo Motors:** Employ feedback control to maintain a specific position or speed.
Application: Steering systems (e.g., electric power steering), camera stabilization for advanced driver assistance systems (ADAS).

5. **Hydraulic, Pneumatics:** Use fluid power to generate mechanical motion; hydraulics use liquids, while pneumatics use compressed air. Application: Brake systems (e.g., ABS, brake booster), suspension systems, and adjustable seats.

Key elements of Mechatronics systems

Sensors are devices that detect and measure physical properties and convert them into electrical signals. Here, sensors play a vital role in providing feedback to control systems. Here are some key types of sensors:

1. Switches: Detect the state of an object or device, either open or closed.
Application: Used in seatbelt sensors, door sensors, headlight on/off switches, and ignition switches to monitor basic vehicle states.

2. Potentiometer: Measures the position or displacement of a rotating shaft, providing variable resistance.
Application: Used in throttle position sensors, volume control in infotainment systems, and position sensing for adjustable seats or mirrors.

3. Photoelectrics: Use light beams to detect the presence or absence of an object.
Application: Widely employed in Parking sensors, adaptive headlights.

4. Digital Encoder: Measures the position, rotation, or angular displacement in a digital format.
Application: Used in Power steering, wheel speed sensors (ABS).

5. Strain Gauge: Measures the deformation or strain in an object subjected to force.
Application: Commonly used in Load cells (tire pressure), structural stress monitoring.

6. Thermocouple: Measures temperature by exploiting the voltage produced at the junction of two different metals.
Application: Widely used in Engine, exhaust, and catalytic converter temperature monitoring.

7. Accelerometer: Measures acceleration forces in one or more directions.
Application: Found in Airbags, stability control, impact detection.

8. MEMS (Micro-Electro-Mechanical Systems): Miniature devices combining mechanical and electronic components on a single chip.
Application: Used in Accelerometers, gyroscopes, tire pressure monitoring.

Key elements of Mechatronics systems

Input signal conditioning and interfacing are essential aspects of mechatronic systems to ensure that signals from sensors or other sources are suitable for further processing. Here's a brief overview of the mentioned elements:

1. Discrete Circuits: Utilized for basic signal processing and manipulation using individual electronic components.

Application: Used in custom circuits for sensor signal processing, such as for fuel level sensors, temperature sensors, or seatbelt warning systems.

2. Amplifiers: Increase the amplitude of a signal, making it stronger for better detection.

Application: Boosting weak signals from sensors like oxygen sensors, crankshaft position sensors, or knock sensors for improved detection by the ECU.

3. Filters: Selectively allow certain frequencies to pass while attenuating others.

Application: Filtering noise from sensors like wheel speed sensors (ABS), GPS receivers, or cabin temperature sensors for more accurate readings. Output signal conditioning and interfacing are crucial for converting processed signals into a form suitable for driving actuators, devices, or other external components. Here's an overview of the mentioned elements:

4. PWM (Pulse Width Modulation): Modulates the width of pulses in a digital signal to simulate an analog signal.

Application: Controlling motor speed for fans, electric power steering, or regulating fuel injectors.

5. Power Transistors: Amplify and control power signals, switching them on or off as needed.

Application: Driving high-power components such as window motors, headlight systems, or electric seats.

6. Power Amplifiers: Amplify the power of signals to drive larger loads or devices.

Application: Amplifying control signals to drive larger loads, like HVAC blower motors, electric actuators in suspension systems, or power windows.

Key elements of Mechatronics systems

Digital control architectures in mechatronic systems encompass various components and technologies for effective system operation. Here's a breakdown of the mentioned elements:

1. Logic Circuits: Basic building blocks that perform logical operations (AND, OR, NOT) using digital signals.
Application: Decision-making in systems like airbag deployment, light control, or ABS.

2. Microcontroller: Integrated circuit containing a processor core, memory, and peripheral interfaces for controlling and processing data.
Application: Central control for engine management systems, powertrain control, and infotainment..

3. SBC (Single Board Computer): Complete computer system on a single board, typically including a processor, memory.
Application: Advanced computing tasks in autonomous vehicles, infotainment systems, and navigation.

4. PLC (Programmable Logic Controller): Industrial digital computer specifically designed for control of manufacturing processes.
Application: Used in manufacturing automation for assembly line control in car production.

5. Sequencing, Timing: Determine the order and timing of operations within the control system.
Application: Managing engine start-up sequences, timing for fuel injection, and ABS control.

6. Logic, Arithmetic: Perform logical and arithmetic operations, respectively, on digital data.
Application: Processing sensor data for systems like traction control, stability control, and cruise control.

7. Control Algorithms: Sets of instructions and logic that dictate the behavior and response of the mechatronic system.
Application: Used in adaptive cruise control, traction control, and energy management in electric vehicles. performance.

8. Communication: Facilitates data exchange between different components within the mechatronic system.
Application: Data exchange between ECUs (e.g., CAN bus) for coordinated vehicle systems like airbags, navigation, and engine control.

Key elements of Mechatronics systems

User interfaces in mechatronic systems involve both input and output devices to facilitate interaction. Here's an overview of the mentioned elements:

Inputs:

1. Buttons: Provide discrete input by pressing or releasing.
Application: Used for controls like engine start/stop, door locks, or seat adjustments.

2. Keypad, Keyboard: Input devices with keys for alphanumeric input.
Application: For entering PIN codes, radio station presets, or vehicle settings.

3. Joystick, Mouse: Enable multidirectional input for navigation or control.
Application: Used in infotainment systems for navigation and selection, or in advanced driver-assistance systems (ADAS).

4. Microphone: Converts sound waves into electrical signals for audio input.
Application: Voice commands for hands-free calling, navigation, and voice-activated controls.

5. Touch Screen: Allows users to interact by touching the display.
Application: Central control interface for infotainment, climate control, and navigation systems.

Outputs:

1. LEDs (Light-Emitting Diodes): Emit light and are often used for visual indicators.
Application: Indicate vehicle status (e.g., brake light, turn signals, battery status).

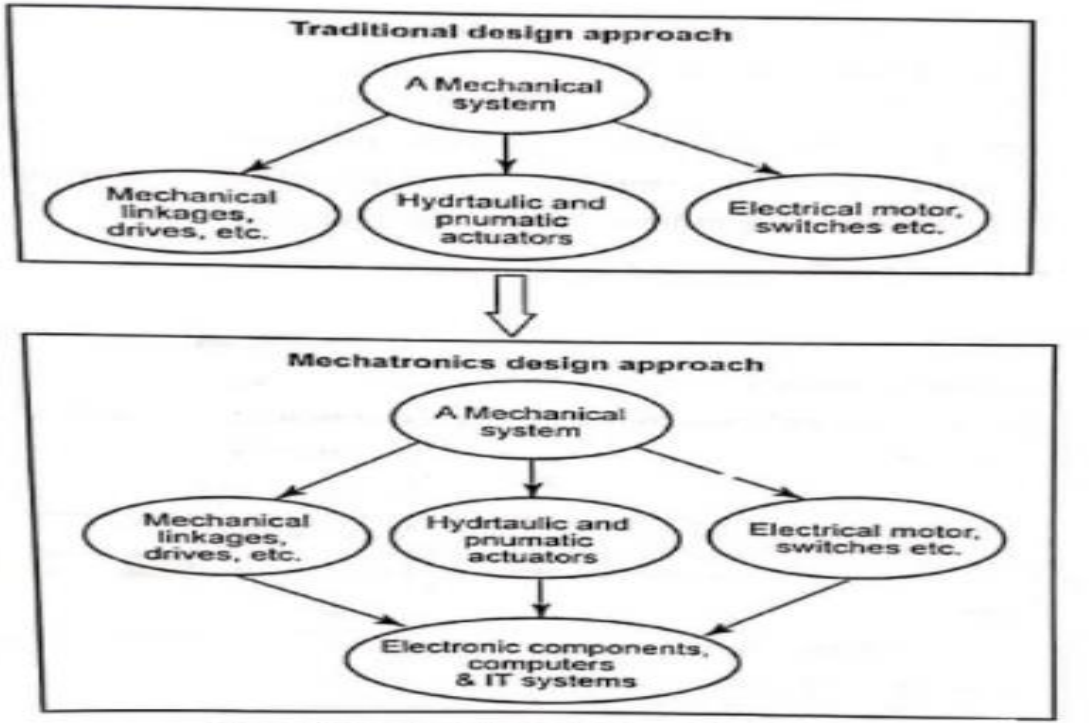
2. Digital Displays: Show numerical or alphanumeric information digitally.
Application: Display vehicle information like speed, fuel levels, and navigation directions

3. LCD (Liquid Crystal Display): Display technology that uses liquid crystals for visual output.
Application: Instrument cluster displays for speed, engine status, and warning messages.

4. Monitor/Screen: Display visual information, often in a larger format than LCDs.
Application: Larger displays for infotainment systems, rear-view cameras, and navigation.

5. Buzzer/Speaker: Produce audible sounds.
Application: Used for warnings, alerts (e.g., seatbelt reminder), and system notifications

Mechatronics design process



Traditional and mechatronics design approaches

Mechatronics design process

Traditional Design Approach

A Mechanical System: *Traditional mechanical systems design tends to prioritize functionality***,** reliability, and manufacturability. Engineers in this framework focus on achieving specific performance metrics and meeting predetermined specifications. However, this approach may sometimes limit innovation and flexibility due to its adherence to established norms and standards.

Mechanical linkages, drives, etc: *Traditional mechanical systems often rely on physical linkages and* drives to transmit motion or force. These systems employ gears, levers, pulleys, and other mechanical linkages to achieve desired functions. Their design emphasizes tangible connections and physical components, offering a hands-on, tangible approach to accomplishing tasks through the transfer and transformation of mechanical energy.

Mechatronics design process

Traditional Design Approach

Hydraulics and Pneumatics as actuators: *Traditional mechanical systems use hydraulics and* pneumatics as actuators. Hydraulics use pressurized fluids to generate motion, offering high force capabilities but often requiring more maintenance. Pneumatics, on the other hand, employ compressed air, providing speed and simplicity, albeit with slightly lower force. Both systems have been foundational in traditional designs, offering distinct advantages and trade-offs in various applications.

Electrical motors, switches, etc: *Traditional mechanical systems employ a range of electrical* components such as motors, switches, and relays to facilitate their operation. Motors serve as the driving force, converting electrical energy into mechanical motion. Switches and relays control the flow of electricity, enabling functionalities like on/off operations or directing power to specific components. This classic approach, while effective, often involves simpler designs compared to modern electronic systems, relying on these fundamental electrical elements to power and regulate mechanical functions.

Mechatronics design process

Mechatronics Design Approach

Mechatronics design is an interdisciplinary approach that integrates mechanical engineering, electronics, computer science, and control engineering to create intelligent systems. It emphasizes a holistic perspective, combining these diverse fields to design and develop efficient, adaptable, and multifunctional systems. The approach aims to optimize the performance, functionality, and reliability of machines by merging mechanical components with sensors, actuators, and computer control, resulting in innovative solutions across various industries.

Mechanical linkages, Drives, etc: Mechatronics design employs various mechanical linkages and drives to integrate mechanical systems with electronics and controls. These elements form the backbone of mechatronic systems, enabling precise motion, force transmission, and control. Linkages like gears, belts, and cams translate motion, while actuators and motors drive these mechanisms, allowing for dynamic functionality. Efficient design and integration of these components are crucial in achieving seamless interaction between mechanical and electronic elements, fundamental to the success of mechatronic systems.

Mechatronics design process

Mechatronics Design Approach

Hydraulic and pneumatic actuators: Hydraulic and pneumatic actuators are pivotal in mechatronics design, offering dynamic control in diverse applications. Hydraulics leverage fluid power to generate force, ideal for heavy-duty tasks due to high force capabilities. Meanwhile, pneumatics utilizes compressed air for lighter loads, offering rapid response times. Both actuators enable precise motion control, making them integral in mechatronics, where efficiency and accuracy are paramount. Their application varies based on specific design requirements, with hydraulics excelling in heavy lifting and pneumatics in agile, high-speed operations.

Electrical Motors, Switches, etc: Mechatronics, at its core, integrates electrical motors, switches, and various components to create intelligent systems. Electrical motors serve as the workhorses, converting electrical energy into mechanical motion. Switches act as the gatekeepers, controlling the flow of electricity within the system. Together, they form the backbone of mechatronic designs, enabling precise control, automation, and the seamless interaction of mechanical and electrical elements.

Electronic Components, Computers, IT systems: In mechatronics design, electronic components form the backbone, integrating with mechanical systems. Computers act as the brain, orchestrating operations, and IT systems facilitate data exchange. Together, they enable the seamless synergy of mechanics, electronics, and computing, pivotal in creating advanced and efficient mechatronic systems.

RV College of Engineering®

Phase of Mechatronic system design process

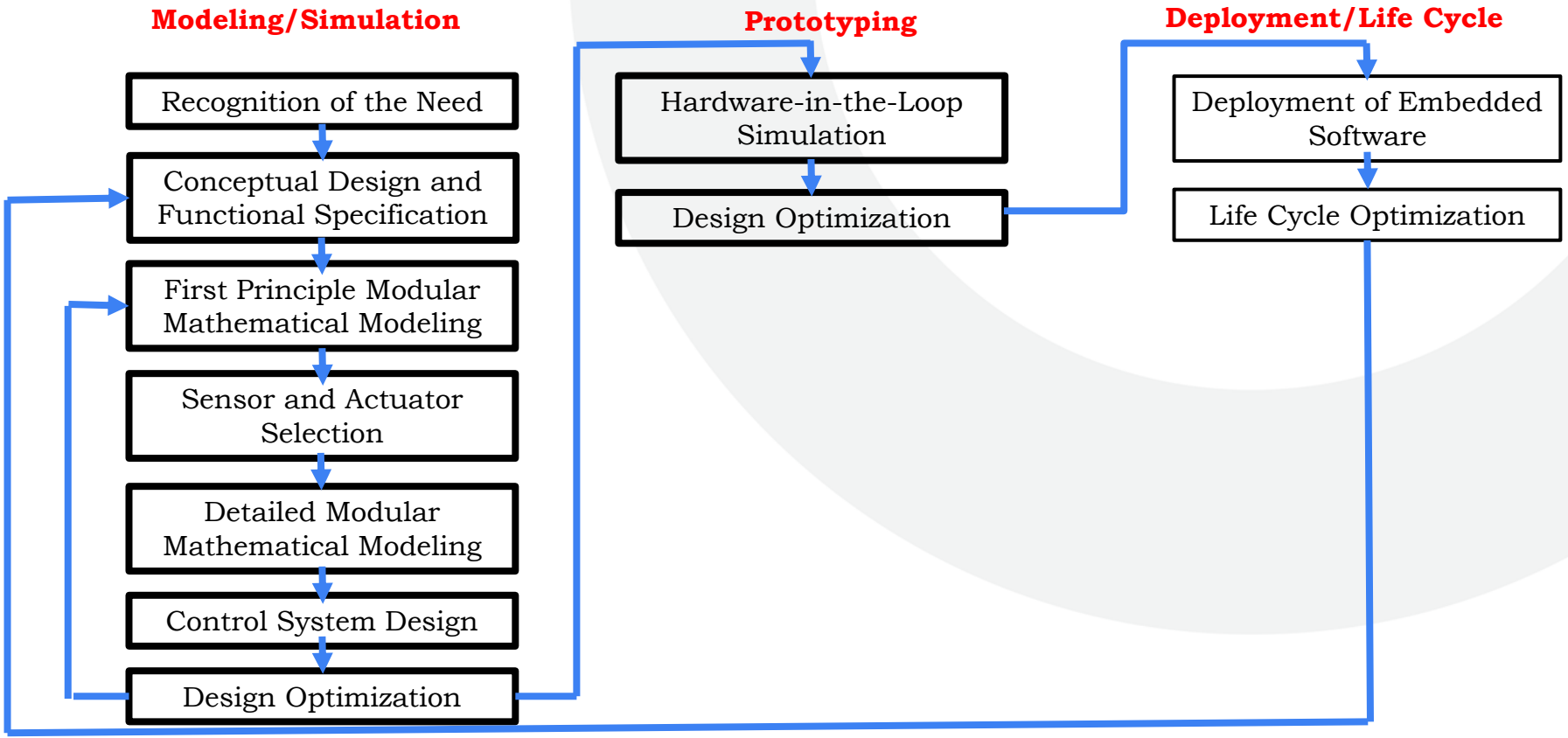
Go, change the world®

- Modelling & Simulation
- Prototyping
- Deployment

RV College of Engineering®

Phase of Mechatronic system design process

Go, change the world®



Phase of Mechatronic system design process

MODELING / SIMULATION

1. Recognition of the need- In the context of simulation or modeling of a car, the recognition of need involves identifying the specific problem or scenario that the simulation of car aims to address. It's the initial step where you define what you want to simulate and why, laying the foundation for the modeling process.

2. Conceptional design and functional specification- Conceptual design of a car involves outlining the high-level structure and key elements of the model. It's about defining the overall framework of a car before delving into specifics. Functional specification, on the other hand, details how the model of a car will operate, specifying its functions, features, and interactions. Together, these steps provide a roadmap for developing a detailed and effective model and simulation.

3. First-principle modular mathematical modeling- With respect to a car. it refers to a systematic approach where a complex system is broken down into fundamental principles or basic components. Each component is represented mathematically based on first principles, which are the fundamental laws or equations governing the system. The modular aspect involves organizing these mathematical representations into modules, making it easier to manage and understand the overall system.

Phase of Mechatronic system design process

MODELING / SIMULATION

4. Sensor and actuator Selection- A sensor is defined as a device in a car that receives a signal or stimulus and responds to it by generating an electrical signal. The output signals correspond to some forms of electrical signal, such as current or voltage, which can be easily measured.

5. Detailed Modular Mathematical Modeling- Develop mathematical equations based on physics principles. - Examples: Forces, torques, displacements.

6. Control System Design- The control system design process of a car involves specifying system requirements, creating a mathematical model, designing a controller to meet those specifications and analyzing system performance through simulations or experiments. It typically follows a cycle of design, analysis, and refinement to achieve desired control system behaviour.

7. Design optimization- in the context of mechatronic system prototyping of a car includes: Iterative Improvements: Refining the system design through successive iterations to enhance performance

Phase of Mechatronic system design process

PROTOTYPING

8. Hardware in Loop Simulation Hardware in Loop (HIL) Simulation: In a car, it is a crucial aspect of the prototyping phase in mechatronic system design. It involves: Integration of Real Hardware: Connecting physical components or subsystems with simulation software.

9. Design Optimization- Design optimization in a car involves defining design objectives and constraints, creating mathematical models to represent the design space, employing optimization algorithms to search for the optimal solution, and iteratively refining the design based on the obtained results.

Phase of Mechatronic system design process

DEPLOYMENT/LIFE CYCLE

10. Deployment of embedded software: The development cycle for embedded software in a car typically involves stages like requirements analysis, design, coding, testing, and debugging. After thorough testing, the software is deployed onto the embedded system, ensuring it meets performance and reliability criteria. The deployment process may vary based on the specific embedded platform and project requirements.

11. life cycle optimization: The development cycle optimization of a car involves refining and streamlining processes throughout the software development life cycle. This includes continuous improvement in areas like planning, coding, testing, and deployment to enhance efficiency, reduce errors, and ensure timely delivery of high-quality software.

Comparison Of Traditional and Mechatronic Design

TRADITIONAL	MECHATRONICS
Sequential approach	Concurrent approach
Process controlled by relay logic	Microprocessor based programmable logic controller
More wiring to control computer and control room	Reduced wiring and machine cycles stored and executed via local control loops
Manual handling processes for loading and unloading	Use of general purpose robotic for handling; automatic tool changing
Maintenance on a preventive or breakdown basis	Based on in – line diagnostics and condition monitoring

Control System

A control system can be thought of as a system which for some particular input or inputs is used to control its output to some particular value, give a particular sequence of events or give an event if certain conditions are met.

Example:

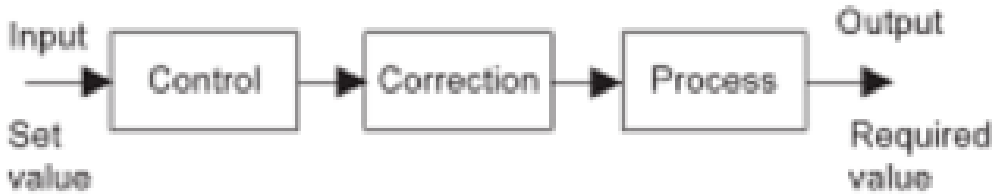
- Central heating system
- Domestic washing machine
- Safety lock system

Two basic forms of control systems are open loop and closed loop

Control System
Open Loop Control System

Basic elements of an open loop control system are:

- Control element
- Correction element
- Process



Example: Automatic toaster system

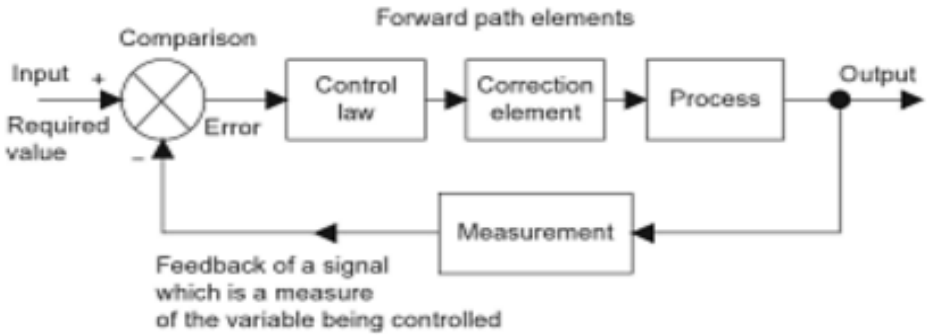
RV College of Engineering®

Control System
Closed Loop Control System

Go, change the world®

•Basic elements of a closed loop control system are:

- Comparison element
- Control law
- Correction element
- Process
- Measurement element



•Example: Automatic water level control, Central heating control

Control System of a Mechatronic System

•The control system for a mechatronic system can be classified as either a discrete event control system or a feedback control system.

•In discrete event system, the controller controls the execution of a sequence of events.

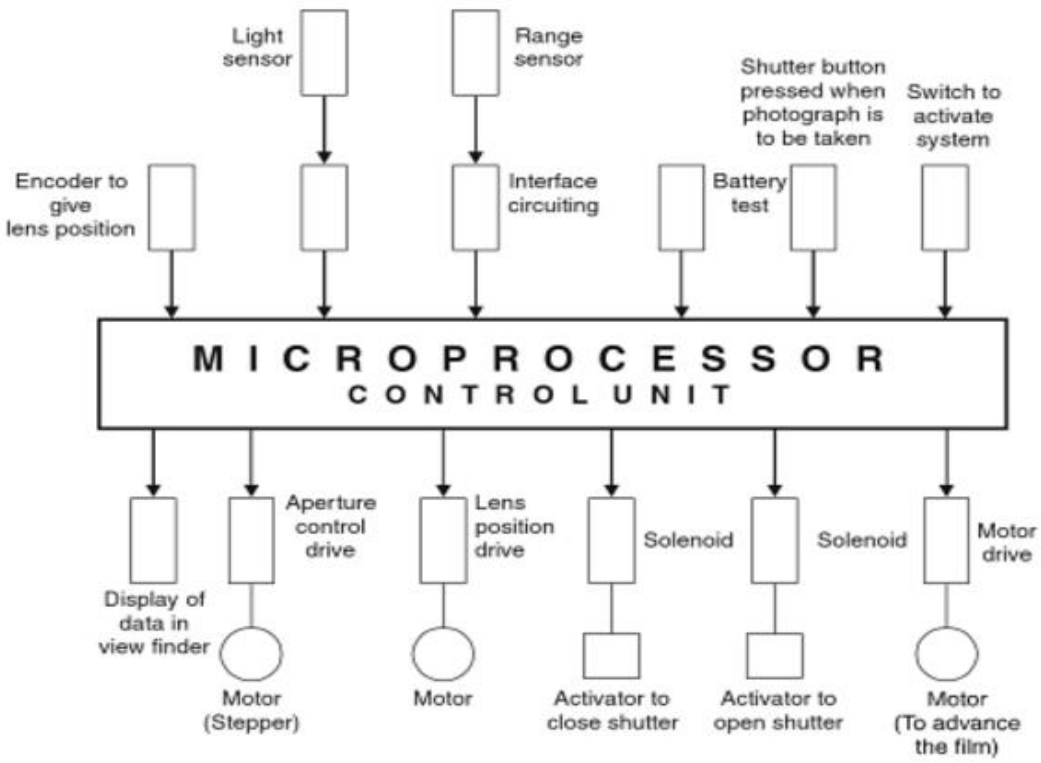
•In a feedback control system, the controller controls one or more variables using feedback sensors and feedback control laws.

•Example: Automatic washing machine, Automatic camera.

RV College of Engineering®

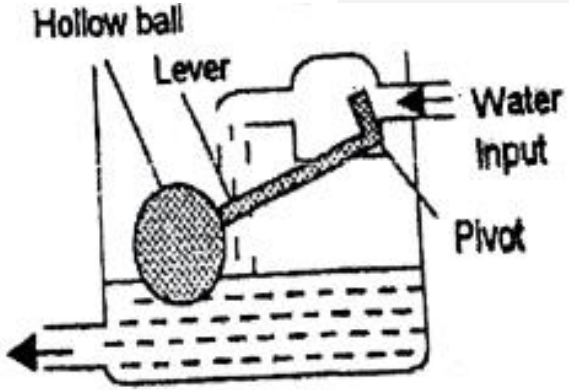
Mechatronic System
Automatic Camera

Go, change the world®



Mechatronic System

Water level



Various elements of controlling the water level in the tank

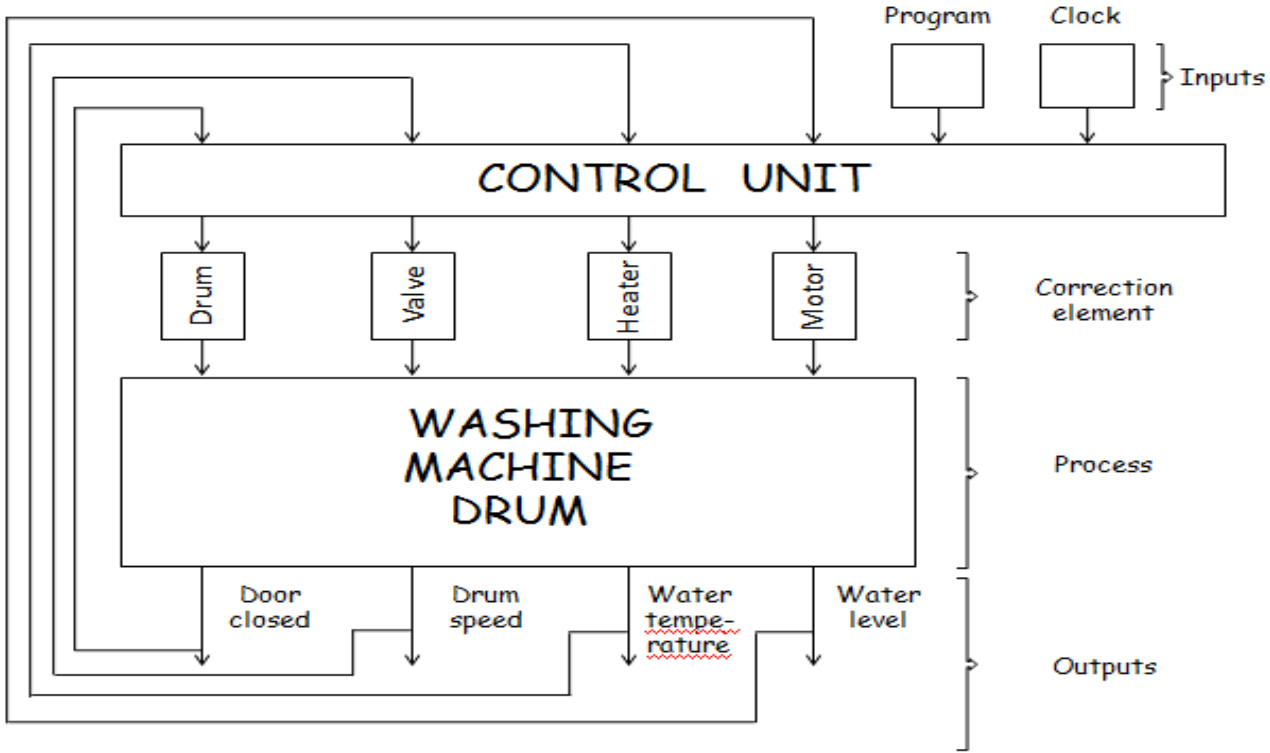
- ✓ Controlled variable – the water level in the tank
- ✓ Reference value – initial setting of the float to the level required
- ✓ Comparison element – the lever
- ✓ Error signal – the difference between the measured and the initial setting of the lever positions
- ✓ Control unit – the pivoted lever
- ✓ Correction unit – the flap opening or closing the water supply
- ✓ Process – the water level in the tank
- ✓ Measuring device – the float ball and the lever

RV College of Engineering®

Mechatronic System

Automatic Washing Machine

Go, change the world®



Mechatronic System

Automatic Washing Machine

Inputs:

1. The program in a washing machine represents the specific set of instructions or a pre-defined sequence of actions that the machine follows during a wash cycle. These instructions dictate the duration, water temperature, agitation pattern, spin speed, and other parameters of the washing process.
2. The clock in this context refers to the timer or timing mechanism within the washing machine. It manages the duration of each phase of the washing cycle.

Correction elements:

1. Drum: It is the part of the machine where clothes are placed for cleaning. The drum's design and function contribute to the effectiveness of washing

2. Valve: valves play a crucial role in controlling the flow of water into and out of the machine. There are generally two types of valves involved: the water inlet valve and the drain valve.

- Water valve:- The water inlet valve is responsible for controlling the flow of water into the washing machine.
- Drain valve:- The drain valve is responsible for removing used water from the washing machine after the washing and rinsing cycles.

3. Heater: the heater in an automatic washing machine contributes significantly to the cleaning efficiency and versatility of the appliance. It allows users to customize their washing experience by selecting the appropriate water temperature for different types of clothes and laundry requirements.

4. Motor: Its primary role is to provide the necessary power to move and control the motion of different parts of the washing machine.

RV College of Engineering®

Mechatronic System

Automatic Washing Machine

Go, change the world®

Process:

1. Loading
2. Detergent Dispensing
3. Water Inlet
4. Agitation
5. Soaking
6. Drainage
7. Rinse Cycle
8. Drainage (Again)
9. Spin Cycle
10. End of cycle

Outputs:

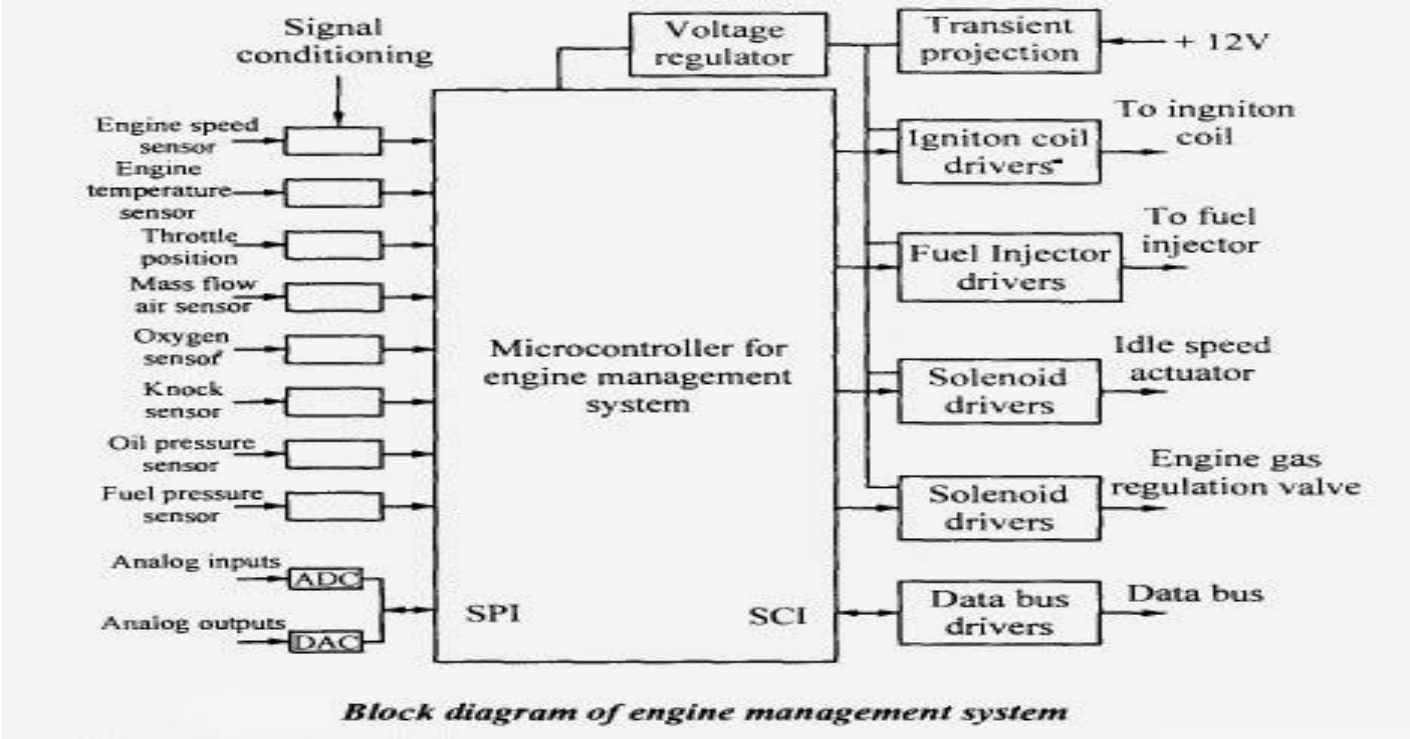
1. Door closed:- the door of the washing machine must be closed because it helps in water containment, user safety ,temperature safety and operational safety.
2. Drum speed:- . The speed of the drum is controlled by the washing machine's motor, and it can vary during different phases of the washing cycle and also the drum speed may be changed depending upon the fabrics that are being washed.
3. Water temperature:- the water temperature in an automatic washing machine is a versatile tool that users can leverage to achieve the desired cleaning results while considering factors such as stain removal, fabric care, energy efficiency, and the specific needs of different types of laundry loads.
4. Water level:- the water level in an automatic washing machine is a key parameter that users can adjust to optimize cleaning performance, water efficiency, and energy efficiency. Maintaining correct water level prevents water leakage and overloading.

RV College of Engineering®

Mechatronic System

Engine Management System

Go, change the world®



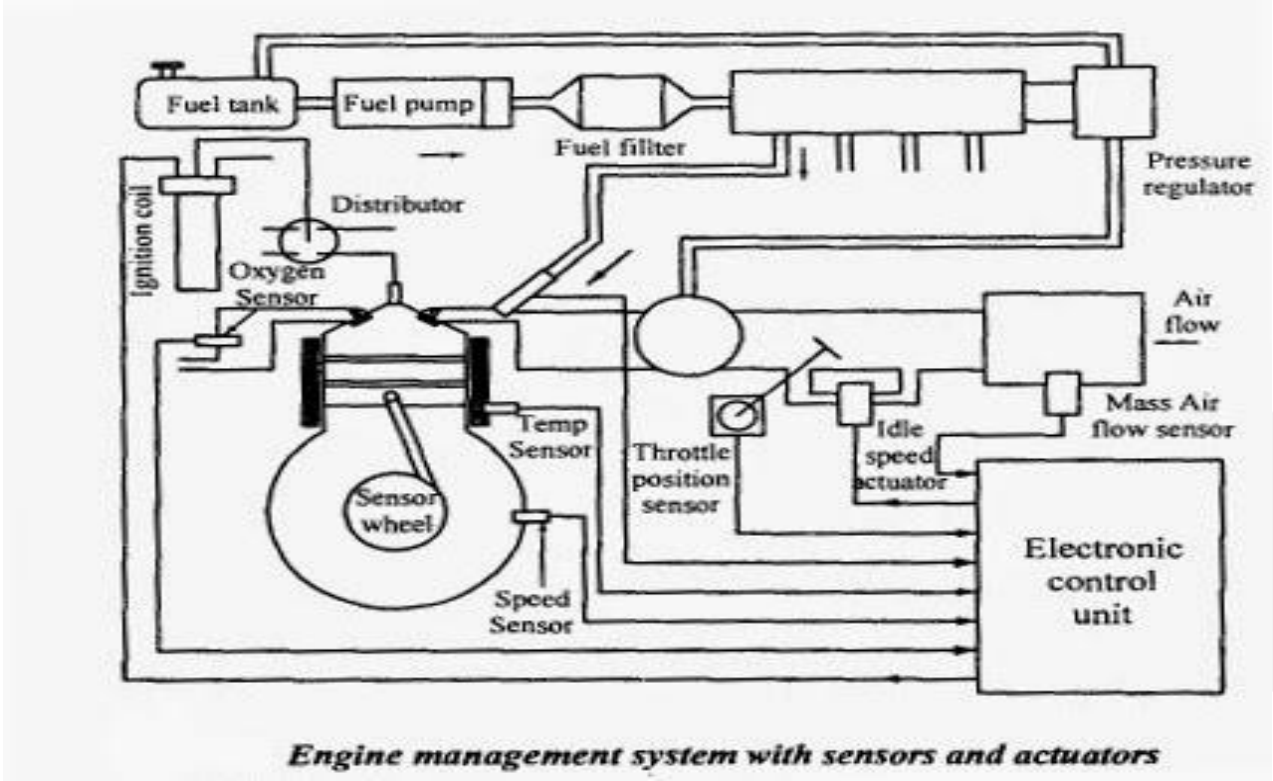
Block diagram of engine management system

RV College of Engineering®

Mechatronic System

Engine Management System

Go, change the world®



Engine management system with sensors and actuators

Mechatronic System

Engine Management System

Microcontroller for Engine Management System(EMS):

- **A microcontroller (MC, µC) or microcontroller unit (MCU)** is a small computer on a single integrated circuit. A microcontroller contains one or more CPUs along with memory and programmable input/output (I/O) peripherals.
- **Central Component:** The microcontroller acts as the brain of the engine management system. It processes the signals from various sensors, makes real-time decisions, and sends commands to the actuators to manage the engine's performance.
- **Functions:** Controls ignition timing, fuel injection, idle speed, and other critical engine functions by processing sensor data



Mechatronic System

Engine Management System

Signal Conditioning:

- In electronics and signal processing, **signal conditioning** is the manipulation of an analog signal in such a way that it meets the requirements of the next stage for further processing.
- **Purpose:** This component processes raw signals from sensors and converts them into a format suitable for the microcontroller to read and interpret.
- **Example:** Amplifying weak sensor signals, filtering noise, and scaling voltage levels for accurate data interpretation.

Mechatronic System

Engine Management System

Sensors:

- **Engine Speed Sensor:** Monitors the rotational speed of the engine (RPM). Operates on the basis of a magnetic pickup or Hall effect. Essential for determining ignition timing and fuel injection rates.
- **Engine Temperature Sensor:** Measures the temperature of the engine coolant to adjust fuel mixture and engine timing for optimal combustion and prevent overheating. Usually a thermistor.



Speed sensor



Temperature sensor

Mechatronic System

Engine Management System

Sensors:

- **Throttle Position Sensor:** Detects the position of the throttle valve to determine how much air is entering the engine. It is used to adjust fuel injection and ignition timing. Typically, this is a potentiometer or a non-contact sensor.
- **Mass Flow Air Sensor:** Measures the amount of air entering the engine to calculate the correct fuel-to-air ratio. It may operate on a hot wire or hot film principle.



Throttle position sensor



Mass air-flow sensor

Mechatronic System

Engine Management System

Sensors:

Oxygen Sensor: Monitors the level of oxygen in the exhaust gases to help maintain the optimal air-to-fuel ratio and reduce emissions. This generates a voltage based on oxygen concentration differences between inlet and exhaust using a zirconia or titanium dioxide element.

- **Knock Sensor:** Detects engine knocking or pinging (abnormal combustion) and helps adjust ignition timing to prevent engine damage. Typically, this is a piezoelectric sensor.



Oxygen sensor



Knock sensor

Mechatronic System

Engine Management System

Sensors:

Oil Pressure Sensor: Measures the oil pressure to ensure proper lubrication of engine parts. Often uses a pressure-sensitive diaphragm connected to an electrical circuit that changes voltage based on pressure levels.

- **Fuel Pressure Sensor:** Monitors the pressure of the fuel in the fuel line to maintain proper fuel delivery. Typically uses a diaphragm or strain gauge which reacts to pressure changes and converts them into electrical signals.



Oil pressure sensor



Fuel pressure sensor

Mechatronic System

Engine Management System

Voltage Regulator:

- A **voltage regulator** is a system designed to automatically maintain a constant voltage. It may use a simple feed-forward design or may include negative feedback. It may use an electromechanical mechanism, or electronic components.
- **Purpose:** Ensures that a constant voltage level is supplied to the microcontroller and other components for reliable operation.
- **+12V Supply:** Represents the power supply typically provided by the vehicle's battery or alternator.



Mechatronic System

Engine Management System

Drivers and Actuators:

- **Ignition Coil Drivers:** Control the ignition coil to generate the high voltage needed to create a spark in the engine cylinders for combustion.
- **Fuel Injector Drivers:** Manage the opening and closing of fuel injectors to regulate the amount of fuel sprayed into the combustion chamber.

Solenoid Drivers: Control solenoid-operated devices such as the idle speed actuator and engine gas regulation valve, adjusting the engine's air intake and emissions control. Solenoids are electromagnetic devices that convert electrical energy into mechanical motion.



- **Idle Speed Actuator:** Maintains a steady idle speed by controlling the amount of air that bypasses the throttle when the vehicle is stationary.
- **Engine Gas Regulation Valve:** Adjusts the amount of exhaust gas re-circulated to control emissions and improve fuel efficiency.

ROBOT

The word robot was first coined by Czech novelist KAREL CAPEK in 1920's. The word robot means a servant or a worker. However, this was just a word coined but the actual technological development in the field of robot was first noted in the year 1957 by English inventor CYRIL WALTER KENWARD developed a manipulator that could move in x-y-z axes system.

BASIC ELEMENTS OF ROBOT or ROBOT ANATOMY

The basic elements of industrial robots are manipulator, controller, end effector, sensors and energy source:-

1. Manipulators: It comprises of base, arm, wrist are the most obvious parts of a robot. The robot movements are executed by the mechanical parts like links, power joints and transmission system along with internal sensors housed within the manipulator.

2. Controller: The controller acts like a brain of robot. It performs the functions of storing and sequencing data in memory, initiating and stopping the motions of manipulators and interacting with environment.

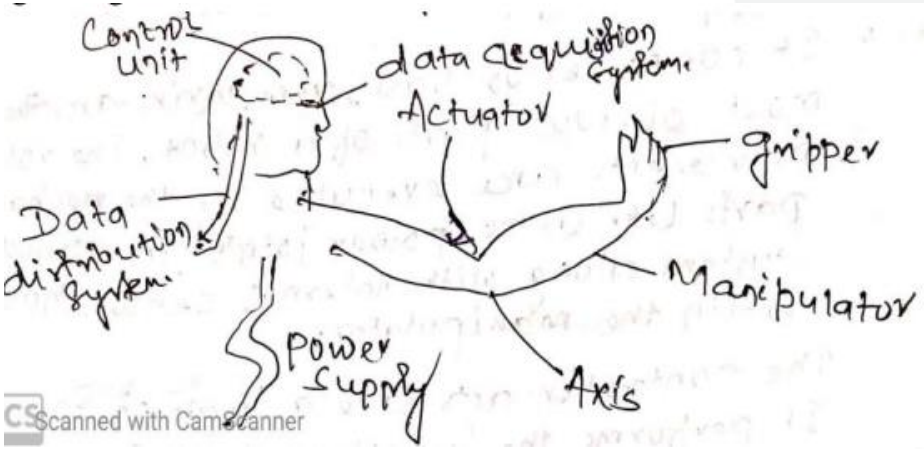
3. End effectors: End effector is the tool, a sort of gripper which directly interacts with the job. Grippers are being designed to handle a wide range of part configurations. The gripper is similar to human hand. Just the hand grasps the tool to perform the work; the gripper secures the robot's work piece while the operation is being performed. The shape of the gripper is determined by the task it has to perform. These include tools designed to weld, paint or perform machining operations such as milling and grinding.

4. Actuators: Just as the human body requires muscles the robot arm requires actuators to move the manipulators. The actuators are devices that exert force to drive the manipulator into a predetermined position or series of positions and hold the joint rigidly once the position is reached. There are two types of actuators. Angular and linear. Angular actuators rotate their loads. Linear actuators extend their loads. The classification of actuators is related to the type of power they use. Electrical actuators use electrical power, hydraulic actuators use hydraulic power and pneumatic actuators use pneumatic power.

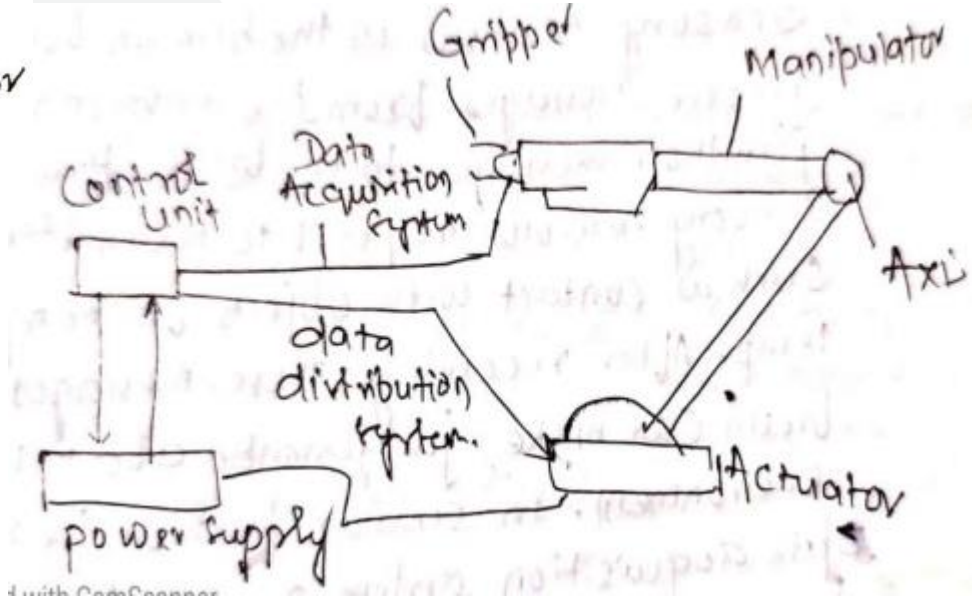
RV College of Engineering®

Go, change the world®

BASIC ELEMENTS OF ROBOT
or
ROBOT ANATOMY



Scanned with CamScanner



BASIC ELEMENTS OF ROBOT
or
ROBOT ANATOMY

5. Axes/Joints/ Degree of freedom: While the manipulator provides the support needed there must be flexible joints in the system to follow for movement in different directions. The axes are flexible joints in the system to allow for movement in different direction. The axes are flexible pivots in the mechanical skeleton that allow the bending of structure at that point.

6. Power supplies: Just as human digestive system converts food into usable energy the robot power supply provides the actuators and control unit with the energy that they need to function. The energy must be in the form that a robot system can use. If the actuators are hydraulic and the workplace only has electrical power available, then the power supply must convert this electrical energy into hydraulic energy.

7. Data distribution system: In the human body neurons receive messages from the brain and passage those messages to the muscles. In the same way the data distribution system receive messages from control unit and passes them on to the actuators.

8. Data acquisition system: Sensory neurons in the human body receive messages from the environment and pass those messages to the brain. Human sensory neurons respond to touch stimuli such as contact with objects or change in temperature. After receiving these messages, the brain can make judgements about the environment. In similar fashion, the robotic data acquisition system receives messages from environment and passes those messages into control unit.

ROBOT

Robot definition: The Robot Institute of America in 1979 defined a Robot as follows:

A Robot is a reprogrammable, multifunctional manipulator designed to move material, parts, tools or specialized devices through variable programmed motions of performance of a variety of tasks.

A robot that is used to perform tasks in industries is called as an "industrial robot". An industrial robot can perform wide range of industrial tasks like loading, unloading, welding, painting, inspection, assembly, material transfer, etc.

Robotics definition: Robotics can be defined "as a field of technology that deals with the conception, design, construction, operation and application of robots".

ROBOT

But why a robot is required when a human can perform the tasks? That is because of the following qualities:

- Robots can work in hazardous or dangerous environments such as a nuclear reactor, behind enemy lines, outside earth's atmosphere, etc. They can substitute humans in such environment.
- Robots can consistently work as per the program with great repeatability and accuracy that is unmatched by the humans.
- At the end of a task, robots can be reprogrammed to take up different task altogether.
- Robot parts can be replaced when damaged.
- In industries, Computer Integrated Manufacturing can be realized by connecting robots to computer system and carrying out the different tasks.

Basic terms related to Industrial Robots

In the study of Robotics, few basic terms are used quote often and their meanings are quite important. The terms used in relation to Industrial Robots are as below:

Manipulator: manipulator is an arm-like mechanism which is designed to manipulate or move materials, parts or tools without direct human contact.

Joint: A joint is the one that integrates two or more links to provide controlled relative movement between input link and the output link.

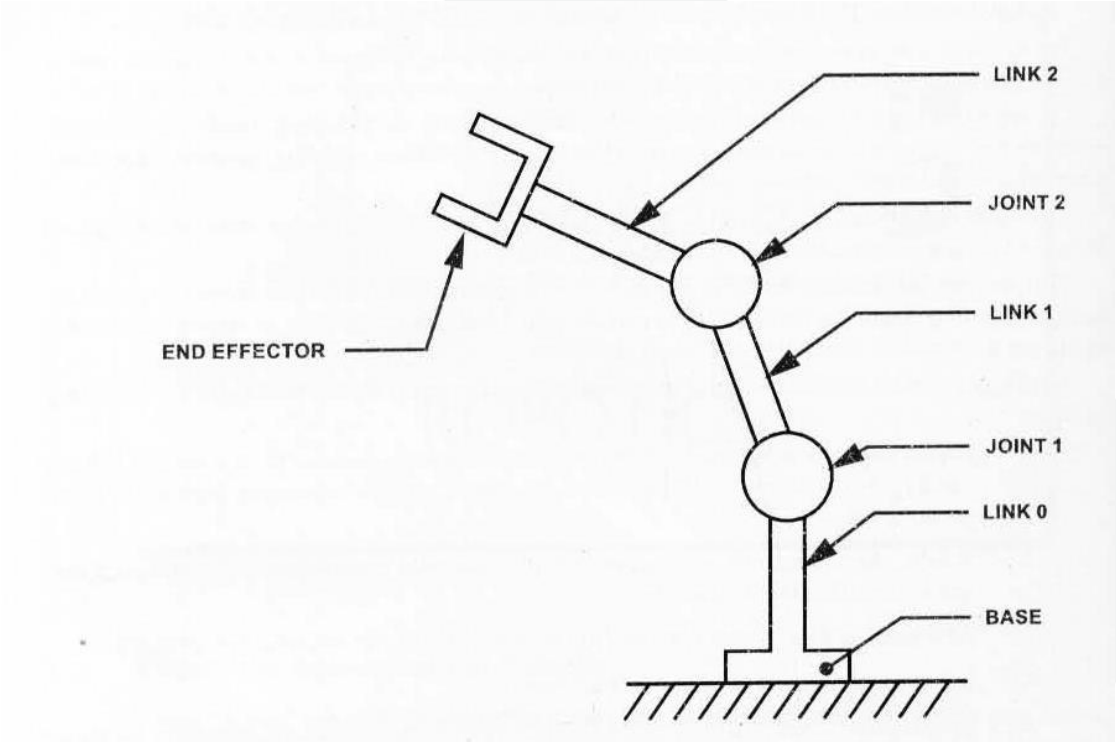
Link: The link is a rigid member that connects the joints. Link can be an input link and an output link. The movement of the input link causes various motions of the output link.

Degrees of freedom (d.o.f): The degrees of freedom describes a robot's freedom of motion in the three-dimensional space.

End effector: End effector or end-of-arm tool is the device at the end of the robotic arm which is shaped like a hand or as a special tool depending upon the application.

Base: The support for the robot arm is called as the base.

ROBOT



Elements of a Robotic System

A Robotic system has the following basic elements:

1. The Robot: This consists of

- The Manipulator which includes the base and the arm assembly.
- End-of-the-arm tooling which is the end-effector.
- Actuators which convert stored energy into movement. Common actuators include electric motors and linear actuators.
- Transmission elements such as ball screws, pulleys, belts, gears, etc.

2. Control System: The control system generates the required signals to co-ordinate and execute the robot movements. The Control System comprises of:

- Controls such as Mechanical control, hydraulic control, pneumatic control, electrical or electronic control. The control techniques can be an open-loop (non-servo) control, feedback control, feed forward control and adaptive control.
- Sensors that allows robots to collect information about a certain measurement of the environment or internal components. The sensors can be touch sensors or vision sensors.
- Equipment Interfaces

3. Computer System: The Computer system is used to program the robots according to the tasks required to be performed. The necessary software must be installed in the computer to develop robot programs.

4. Power Source: Power source supplies electrical energy for the robot. The commonly used power source is the battery which can be a lead-acid battery or a silver-cadmium battery.

Types of Automation

Automation is categorized into three types. They are,

1) Fixed Automation
2) Programmable Automation
3) Flexible Automation.

Fixed Automation:-

It is the automation in which the sequence of processing or assembly operations to be carried out is fixed by the equipment configuration.

In fixed automation, the sequence of operations (which are simple) are integrated in a piece of equipment. Therefore, it is difficult to automate changes in the design of the product.

It is used where high volume of production is required Production rate of fixed automation is high. In this automation, no new products are processed for a given sequence of assembly operations.

Types of Automation

Fixed Automation

Features:-

- High volume of production rates,
- Relatively inflexible in product variety (no new products are produced).
- High initial investment for custom-engineered equipment
- High production rates

Examples:-

- Automobile industries.
- Machining transfer lines and automated assembly machines

Types of Automation

Programmable Automation

It is the automation in which the equipment is designed to accommodate various product configurations in order to change the sequence of operations or assembly operations by means of control program. Different types of programs can be loaded into the equipment to produce products with new configurations (i.e., new products).

It is employed for batch production of low and medium volumes. For each new batch of different configured product, a new control program corresponding to the new product is loaded into the equipment. This automation is relatively economic for small batches of the product

Types of Automation

Programmable Automation

Features:-

- High investment in general purpose,
- Lower production rates than fixed automation,
- Flexibility & Changes in products configuration,
- More suitable for batch production.

Examples:-

- Industrial robot
- NC machines tools

Types of Automation

Flexible Automation

A computer integrated manufacturing system which is a extension of programmable automation is referred as flexible automation. It is developed to minimize the time loss between the changeover of the batch production from one product to other while reloading.

The program to produce new products and changing the physical setup i.e., it 4 produces different products with no loss of time. This automation is more flexible in interconnecting work stations with material handling and storage system

Types of Automation

Flexible Automation

Features:-

- High investment for a custom engineering system.
- Medium Production rates
- Flexibility to deal with product design variation,
- Continuous production of variable mixtures of products.

Example:-

- Flexible manufacturing systems (FMS)

Types of Automation

Advantages of Automation

- High Production rates
- Lead time decreases
- Storing capacity decreases
- Human errors are eliminated.
- Labour cost is decreases.

Dis-Advantages of Automation

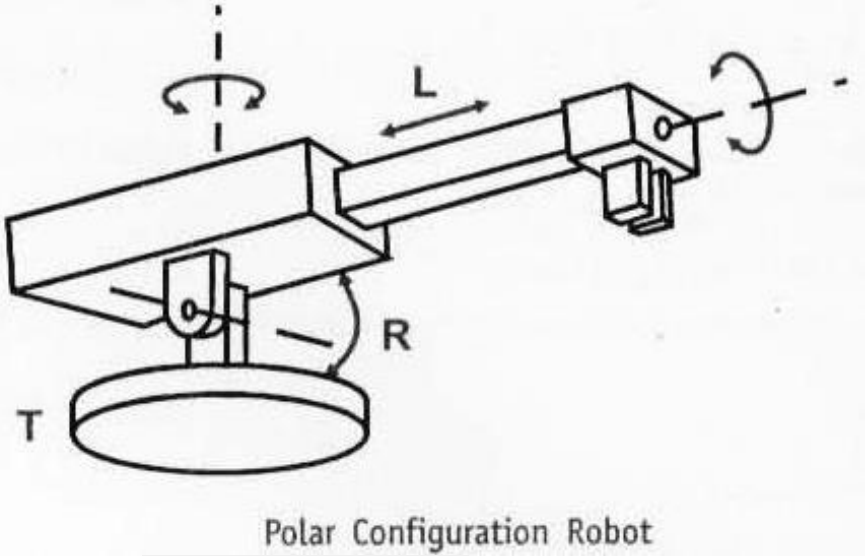
- Initial cost of raw material is very high
- Maintenance cost is high
- Required high skilled Labour
- Indirect cost for research development & programming increases.

Classification of Robot based on Robot configuration:

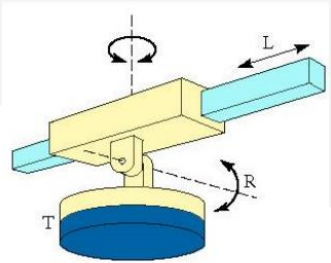
- Polar configuration
- Cylindrical configuration
- Cartesian configuration
- Joint arm configuration

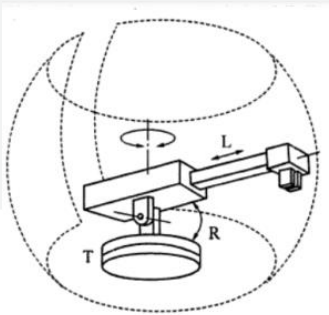
Polar Configuration (Spherical Configuration):

The Polar configuration robots, also called the spherical configuration robots, consist of a sliding arm (L-joint) that is actuated relative to the body and a rotational base along with a pivot, which can rotate about a horizontal axis (R-joint) and the vertical axis (T-Joint). This is shown in the Fig.



Polar Configuration Robot





The one linear and the two rotary joints create a spherical work volume in which the robot operates.
Example: The Unimate 2000 series robot. *(Courtesy: Unimation Inc.)*

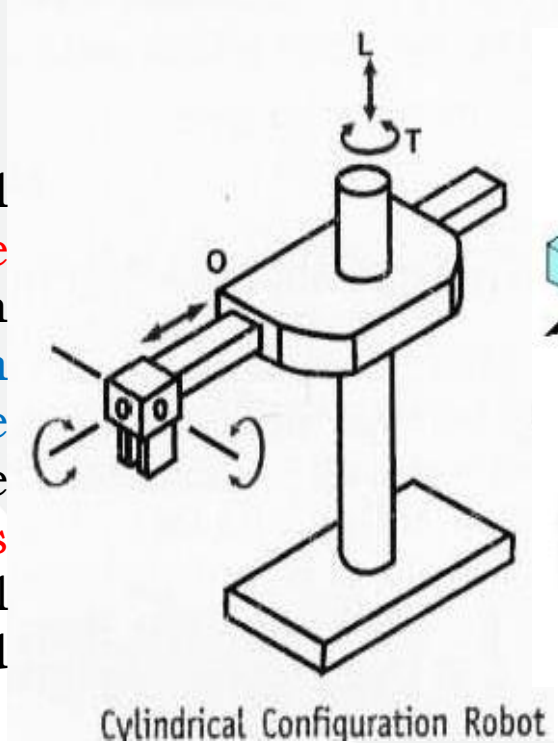
Advantage:
Long reach capability is realized in the horizontal position.

Disadvantage:
The vertical reach is low.

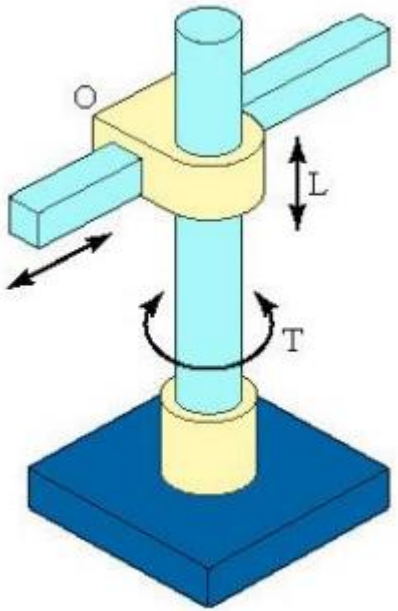
Applications:
Die casting, forging, injection molding, dip coating, cleaning of parts, etc.

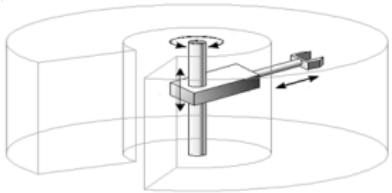
Cylindrical Configuration:

Robots of the cylindrical configuration consist of a slide in the horizontal position and a column in the vertical position. The arm assembly moves up or down relative to the column using an L-joint. The column is rotated about its axis using the T-joint. The radial movement of the arm is achieved using the O-joint, as shown in Fig.



Cylindrical Configuration Robot





Example: Model 1A Robot of GMF Robotics Corp. (Courtesy: GMF Robotics Corp.)

Advantages:

1. Rigidity is increased and is quite robust.
2. Has the capacity to carry high payloads.

Disadvantages:

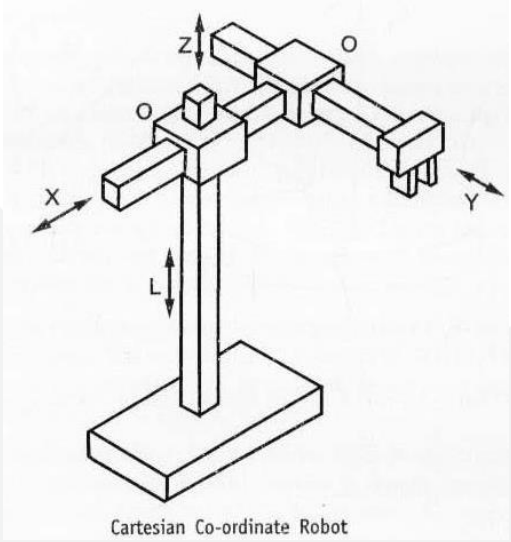
1. Work volume is less.
2. Occupies more floor space.

Applications:

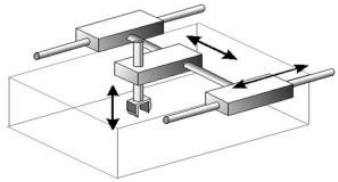
Foundry and forging applications, investment casting, conveyor pallet transfers, machine loading and unloading

Cartesian Co-ordinate Robot:

It is also called as a rectilinear robot or a XYZ Robot. It consists of three sliding joints along the X, Y and Z directions in three dimensional space. There are two orthogonal joints. Since movement can stop and start simultaneously along the X, Y and Z axes, the motion of the tool tip is smoother. A typical Cartesian co-ordinate Robot is shown in the Fig.



Cartesian Co-ordinate Robot



Advantages:

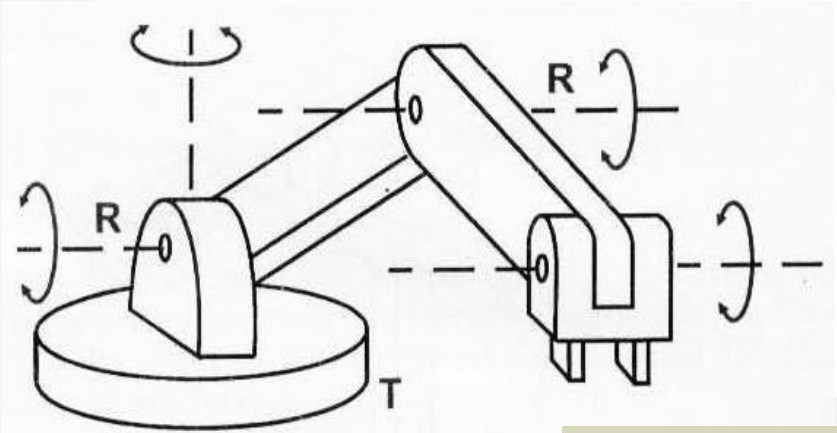
- Allows for simpler controls.
- Possess a high degree of mechanical rigidity, accuracy and repeatability.
- They can carry heavy loads and the weight lifting capacity do not vary within the work envelope.

Disadvantages:

- Limited in their movement to a small and rectangular work space.
- Reduced flexibility.
- Applications:
- To perform pick and place tasks, material handling, loading/unloading and machining operations

Jointed-arm Configuration Robot:

This type of Configuration resembles the human arm where the column swivels about a base (the column and the base forms a T-joint), the column top connects to the shoulder through a shoulder joint (which is the R-joint) and the shoulder connecting to the elbow through an elbow joint (which is also an R-joint). Thus, this configuration has the capability to be controlled at any adjustments in the work space. This is shown in Fig.



Jointed arm Configuration Robot



ROBOT

Advantages:

- The work volume available is large.
- Operation is quick.
- Flexibility is increased.

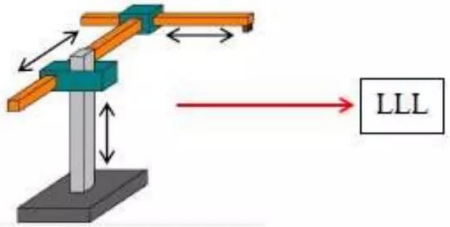
Disadvantages:

- Operating procedures are difficult.
- Quite expensive type of configuration
- Number of components involved are more.

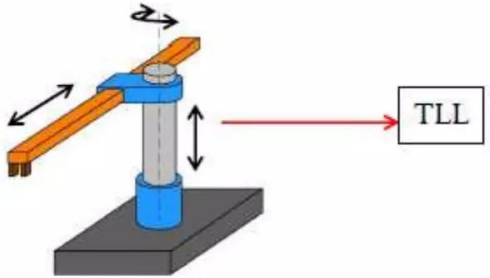
Applications:

- To perform arc welding, spot welding and spray painting operations

Joint Notation Scheme



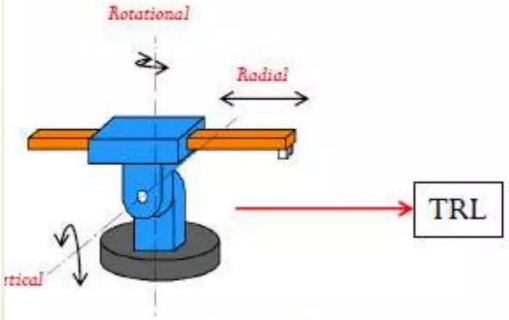
Cartesian coordinate robot



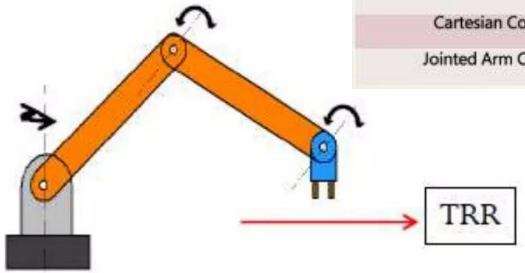
Cylindrical coordinate robot

L-Linear Joint
T-Twisting Joint
R-Rotational Joint
V-Revolving Joint

Robot Configuration (Arm and Body)	Symbol
Polar Configuration	TRL
Cylindrical Configuration	TLL,LTL,LVL
Cartesian Configuration	LLL
Jointed Arm Configuration	TRR, VVR



Polar coordinate robot



Jointed arm robot

Applications of Robot

Material/Part Handling applications

Most of the applications of the Robot falls under this category. Here the Robots are used to Transfer materials from one location to another location. Typical applications are as given below:

a. Pick-and-place application: The most common and simplest application here is the pick-and-place application where objects are picked from one location and placed to another location.

b. Palletizing application: Another application that is slightly complex is the "Palletizing application" where the robot stacks products or carton boxes onto a pallet at various positions on the pallet to the required height. ***Example*** is the KUKA model KR 40 PA that can stack pallets upto 1.6 metre height and that has a payload carrying capacity of 40 kg (Courtesy of Kuka Robotics Corporation). The Robot picks from the same location, but is programmed to place at different locations on the pallet in the stacking row and column order. Here we use the teach-and-play method or program to compute the location of carton box by taking into account the center distance between the carton boxes and the box dimensions of length x width x height.

c. Depalletizing application: Here the robot picks parts from an orderly stacked pallet to another location. ***Example:*** FANUC Robotics M-410iB/140H can handle 140 kg payload and can palletize and depalletize boxes and bags (Courtesy: Fanuc Corporation).

d. Stacking application: Here, the robots are used to stack parts one upon another. After each placement, the vertical position is re-calculated and the new stacking height is determined. ***Example:*** FANUC M-2iA robot (Courtesy: Fanuc Corporation) was used by an organization to stack stamped parts on conveyors.

e. Insertion operation: Here, the robots are used to insert parts into the compartments or spaces provided in a carton.

Applications of Robot

Machine loading/unloading

a. Machine loading/unloading: In the machine loading/unloading application, a robot is used to move the work parts to or/and from the production machine.
The following three possibilities can be included for the loading/unloading application:

b. Machine loading: In the loading operation, a robot is used to load parts into the production machine, but the unloading of the finished parts after completion of operations of the production machine is done by means other than the use of a robot. Example: This is commonly applied in Press working process.

c. Machine unloading: In the unloading operation, a robot is used to unload the finished parts that comes out of a production machine, but the loading of the parts on to the production machine is done by means other than the use of a robot. Example: Die casting and plastic modelling.

d. Machine loading and unloading: In this operation, a robot is used to both load raw parts into the production machine and unload finished parts from the production machine. Example: Machining operation

Applications of Robot

Machine loading/unloading

Following are the industrial applications of robots used for machine loading/unloading:

Die-casting: Here the robot is used to safely unload the parts from a die-casting machine with the safety gates closed. Other connected applications are carrying out cooling tasks or parting agent applications.

Forging: Forging is one of the toughest environments considering the extreme heat, pollution and noise. The use of a robot immensely helps to face the environment of forging. Here the robot loads the red hot billet on to the die of the forging hammer, holds it during the blows and unloads to a safe place away from the hammer.

Plastic injection moulding: Here a robot unloads parts from the injection moulding machine, cuts the runner and drops runner to scrap area.

Sheet metal press operation (Press working): Here a robot loads a blank into the press, then after the press stamping operation is performed the robot unloads the scrap and throws it into the scrap area. The stamped parts from the blank falls in the container placed at the back of the machine.

Machining operations: Here the robot loads the raw blanks on to the machine tool and unloads the finished parts.

Heat treating: Here the robot loads/unloads parts to/from a furnace.

Applications of Robot

Processing operations

Robots are used to carry out the processing operations such as spray painting, spot welding, etc by using a tool at its end-effector. The tools can be a spray painting gun for spray painting operation, a spot welding gun for spot welding operation, etc.

In processing operations, the robot manipulates a tool to perform a process on the work part.

Following are few processes that use industrial robots:

a. Spot Welding: Spot Welding is a metal joining process where two metal parts are squeezed together between two electrodes and subjected to a large amount of current to form a joint at particular points or spots. The end-effector of the robot here is the spot welding gun that applies the approximate pressure and current to the sheet parts to be welded. The spot welding robots have enough number of axes of motion to approach points in the work envelope at any angle. This was difficult to realize in a manned environment in the absence of robots. Spot welding is used largely in the automobile industry to weld automobile bodies such as car panels.

b. Arc Welding: Arc welding is a metal joining process where metals are welded by using the heat generated by an electric arc. The welding here is continuous unlike spot welding. When arc welding is manually carried out, the conditions are difficult for the operators since they require Personal Protective Equipment like welding shield with special glass to avoid UV rays, are under the danger of operating at high temperatures involving high amount of heat and moreover they must be accurate in following the welding path. These problems are now overcome with the use of arc welding robots.

Applications of Robot

Processing operations

The typical components of an arc welding cell includes:

Arc welding robot: Most popular is the jointed-arm type robot because it allows the welding torch to be manipulated similar to a manual operator. The torch and travel angles can be changed to make welds of good quality. Also welding can be done in areas that are difficult to reach. These robots typically consist of five or six free programmable axes.

Power Source: This provides electric power required to do the arc welding operation.

Welding torch: This is carried by the robot as its end-effector tool to direct the welding electrode into the arc.

Wire feeder: These are used to add the filler material in robotic welding.

Welding fixtures: These hold the individual parts to be welded in proper alignment during welding.

Applications of Robot

Processing operations

C. Spray Coating: Spray coating is a process where parts are coated by a spray gun spraying the fluid on to the surface of the part. The fluid passes through the nozzle of the spray gun and is dispersed at high velocity to the surface to be coated. Common examples are powder coating and spray painting.

When human operators carry out the spray coating operations, it results in various health hazards such as the paint or powder fumes / particles entering the operator's body and causing illness, flash fires resulting in burning in the painting booth pit, ear damage due to continued noise of the nozzle, etc. Hence if affordable, robots can be the right substitution to work in such environments.

The robot end gripper is a spray coating gun which sprays the paint or coat on the work surface uniformly. The motion sequence must be smooth with continuous path control. Spray coating robots are used in coating of the exterior and interior parts of a car, spray staining of wood furniture for carousel presentation, applying stain to textiles (like Jeans material), spraying porcelain coating on bathroom fixtures, etc.

Benefits of spray coating robot includes:

1. Protecting human operators from hazardous coating environments
2. Consistency across coated parts is high
3. Better productivity
4. Uniform coating on the coated surface 5. Reduced paint wastage

Applications of Robot

Other Processing applications

a. Grinding: Here the end effector of the robot is the rotating spindle and the grinding wheel is mounted in the spindle chuck. The robot moves its arms relative to the workpiece and grinds the material of the workpiece. Manual grinding has problems like exposure of the operator to fast moving tiny ground particles leading to damage to the eyes and face, requirement of intensive training and grind mark patches. On the other hand, use of robots can give us the perfect grinding finish with saving of time and absence of hazards. As a connected application, robots are used to polish surfaces using a polishing wheel.

b. Drilling and other machining processes: The end-effector is the rotating spindle and the cutting tool is mounted in the spindle chuck.

Drilling a hole manually requires significant time and effort. Using robots in drilling helps to monitor drill speed, torque, countersinking, displacement and breaking of bits. The monitoring is done by sensors on the robots.

Aerospace industry uses robotic drilling due to the precision and productivity benefits. The only challenge of using robots in drilling and other machining operations is that the robot must handle the high cutting forces during machining.

Robots are now used in other machining processes like routing, milling, etc. The end-effector here are spindles on which appropriate tools are mounted according to these machining processes.

Applications of Robot

Other Processing applications

c. Plasma Cutting: Plasma cutting is a process that uses high velocity ionized gas called plasma to heat and melt metals. The plasma later blows away the molten material mechanically to cut the workpiece which is usually one inch thick. Use of robots here results in high quality cuts at faster travel speeds. The plasma torch is attached to the robot as its end-effector.

d. Laser Cutting: Laser cutting is a process that uses a high power laser through optics to cut materials. The laser tool as an end-effector is attached to the robot. The result of robotic laser cutting is precision cutting with high repeatability and accuracy.

e. Waterjet Cutting: Waterjet cutting is a process used to cut parts using a very high pressure stream of water. The nozzle ensures the high pressure of the water stream. When used with a robot, this nozzle becomes the end-effector and the arm moves in the programmed paths to achieve high quality cutting. Waterjet robots are used to cut various materials like metals, fabrics, board, plastic, etc. Waterjet robots is used in aerospace, automotive, textile, packaging and food industries.

f. Other applications include wire brushing, deburring, fiberglass cutting, etc.

Advantages of Robot

Following are some important advantages of using Industrial Robots:

- Robots can be substituted for humans to work in hazardous work environments. Example: During arc welding, in foundry environment, in powder coating environment, etc.
- Robots can produce greater quantity in a short span of time with consistency and accuracy that cannot be matched by humans.
- Robots can work at constant speeds without any break which is not possible by humans.
- Robots are capable of lifting heavy loads without getting tired or injured.
- Robots can work in tight spaces where human reach is not possible.
- Robots can be re-programmed with changed tooling to take up a different task after the end of a batch or a production run. In such cases, robots are better than fixed automation.
- Accidents at the workplace is avoided since robots perform the risky jobs which were otherwise done by humans.
- Since Robots are controlled by computers, they can be integrated to other computer systems to realize Computer Integrated Manufacturing (CIM)
- The usage of robots produces lesser or no defective parts and hence saves time of rework and money to the organization.

Disadvantages of Robot

Following are some disadvantages of using Industrial Robots:

- Organizations have to make huge investments to introduce robots at their workplaces.
- Since parts of a robot are made very precisely, their replacement is very difficult and to maintain, it costs huge amount of money.
- To program and setup the robotic systems and robots, and to avoid unnecessary future problems and mishaps, it requires highly skilled technical engineers and programmers which again is a significant cost for the organization.
- Unless the level of the artificial intelligence is highly sophisticated, robots may not be able to respond properly during times of emergency, during times of accidents or when an unexpected variance occurs.